Karhunen-Loève transform

signal processing community, PCA is also known as the *Karhunen-Loève transform*. In this chapter, we derive PCA from first principles, drawing on our understanding of basis and basis change (Sections 2.6.1 and 2.7.2), projections (Section 3.8), eigenvalues (Section 4.2), Gaussian distributions (Section 6.5), and constrained optimization (Section 7.2).

Dimensionality reduction generally exploits a property of high-dimensional data (e.g., images) that it often lies on a low-dimensional subspace. Figure 10.1 gives an illustrative example in two dimensions. Although the data in Figure 10.1(a) does not quite lie on a line, the data does not vary much in the x_2-direction, so that we can express it as if it were on a line – with nearly no loss; see Figure 10.1(b). To describe the data in Figure 10.1(b), only the x_1-coordinate is required, and the data lies in a one-dimensional subspace of $\mathbb{R}^2$.

10.1 Problem Setting

In PCA, we are interested in finding projections $\tilde{\boldsymbol{x}}_n$ of data points $\boldsymbol{x}_n$ that are as similar to the original data points as possible, but which have a significantly lower intrinsic dimensionality. Figure 10.1 gives an illustration of what this could look like.

More concretely, we consider an i.i.d. dataset $\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_N\}$, $\boldsymbol{x}_n \in \mathbb{R}^D$, with mean $\mathbf{0}$ that possesses the *data covariance matrix* (6.42)

data covariance matrix

$$\boldsymbol{S} = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n \boldsymbol{x}_n^\top . \tag{10.1}$$

Furthermore, we assume there exists a low-dimensional compressed representation (code)

$$\boldsymbol{z}_n = \boldsymbol{B}^\top \boldsymbol{x}_n \in \mathbb{R}^M \tag{10.2}$$

of $\boldsymbol{x}_n$, where we define the projection matrix

$$\boldsymbol{B} := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_M] \in \mathbb{R}^{D \times M} . \tag{10.3}$$

The columns $\boldsymbol{b}_1, \dots, \boldsymbol{b}_M$ of $\boldsymbol{B}$ form a basis of the M-dimensional subspace in which the projected data $\tilde{\boldsymbol{x}} = \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x} \in \mathbb{R}^D$ live.

We assume that the columns of $\boldsymbol{B}$ are orthonormal (Definition 3.7) so that $\boldsymbol{b}_i^\top \boldsymbol{b}_j = 0$ if and only if $i \neq j$ and $\boldsymbol{b}_i^\top \boldsymbol{b}_i = 1$. We seek an M-dimensional subspace $U \subseteq \mathbb{R}^D$, $\dim(U) = M < D$ onto which we project the data. We denote the projected data by $\tilde{\boldsymbol{x}}_n \in U$, and their coordinates (with respect to the basis vectors $\boldsymbol{b}_1, \dots, \boldsymbol{b}_M$ of U) by $\boldsymbol{z}_n$. Our aim is to find projections $\tilde{\boldsymbol{x}}_n \in \mathbb{R}^D$ (or equivalently the codes $\boldsymbol{z}_n$ and the basis vectors $\boldsymbol{b}_1, \dots, \boldsymbol{b}_M$) so that they are as similar to the original data $\boldsymbol{x}_n$ and minimize the loss due to compression.

Example 10.1 (Coordinate Representation/Code)
Consider $\mathbb{R}^2$ with the canonical basis $\boldsymbol{e}_1 = [1, 0]^\top$, $\boldsymbol{e}_2 = [0, 1]^\top$. From